

1 FINAL-STATE STABILITY DOES NOT GUARANTEE A SAFE TRANSITION

SG retirement changes a portfolio one unit at a time. Stability belongs to the whole portfolio, so a stable final state cannot certify every implementation order.

$$\alpha_{\perp}(S, \theta) = \max \operatorname{Re} \lambda(A_{\perp}(S, \theta))$$
$$\mathcal{H}(\theta) = \operatorname{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Each hyperedge is an inclusion-minimal unstable replacement coalition.

synchronous
generators

retire and replace in stages →

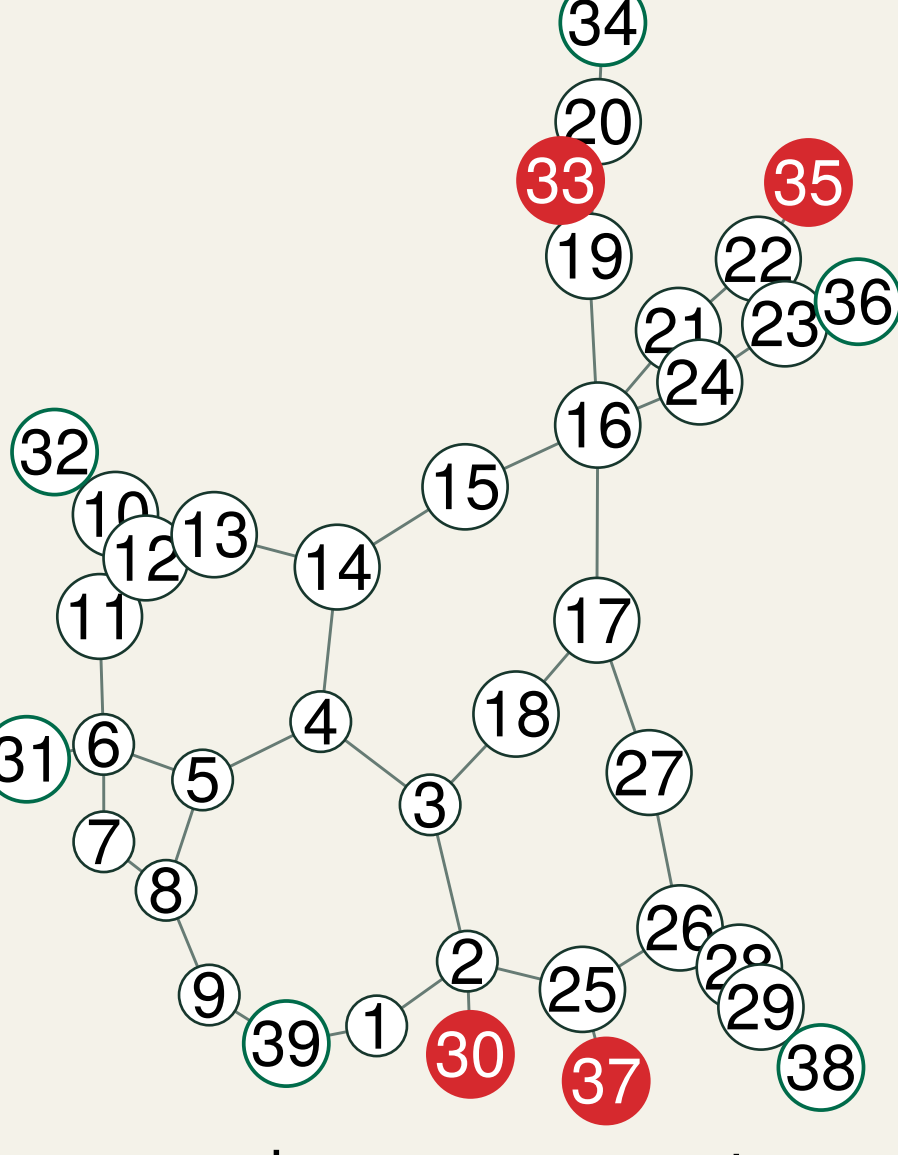
IBR

inverter-based
resources

A stable target can still be unreachable without crossing an unstable portfolio.

2 15/15 SMALLER PORTFOLIOS ARE STABLE; THE COMPLETE PORTFOLIO IS NOT

IEEE-39 one-line
diagram

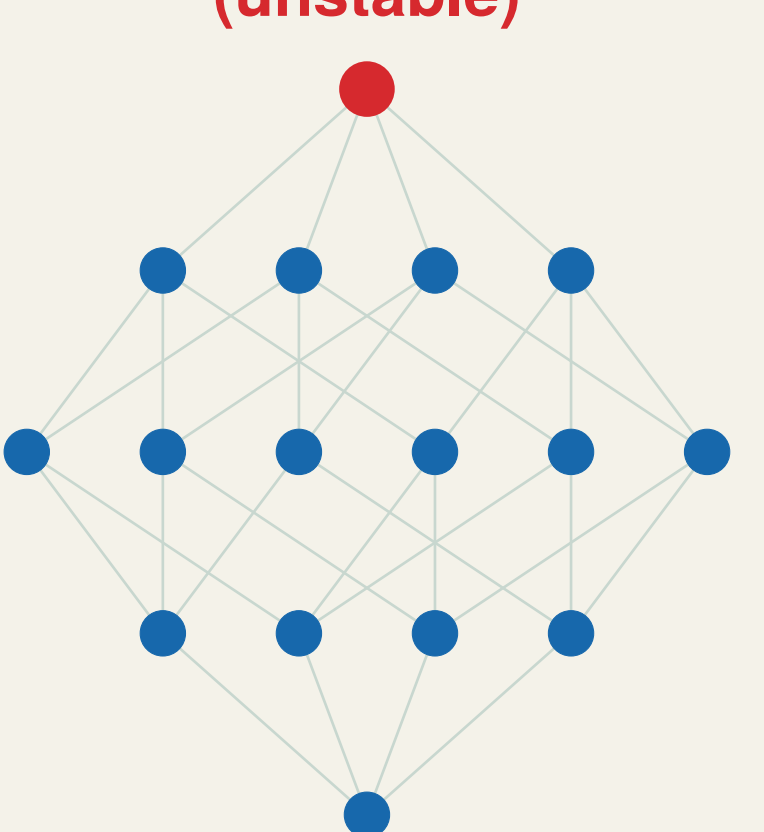


- synchronous generator
- highlighted IBR candidate

15/15
proper subsets stable

$H_4 = \{30, 33, 35, 37\}$
canonical, not universal
P4 control settings
 $g = 0.03625 \quad k = 1.425$
 $t = 1.5 \quad h = 1$
 $\alpha_{\perp} = +0.127 \text{ s}^{-1}$
 $f = 0.6223 \text{ Hz}$
Displaced: 2096.6 MW
Converter: 4270.7 MVA

Hasse lattice ($n = 4$)



- proper subset coalition
- full coalition

H_4 is a minimal instability hyperedge.

3 STABLE TARGET ≠ SAFE PATH

Three levels of safety and how they relate:

A
final-state
stable
 $\alpha_{\perp}(S, \theta) < 0$

B
some safe
order
 $\exists$ all prefixes
stable

C
every order
safe
 $\forall$ orders safe

$C \Rightarrow B \Rightarrow A$
The converse does not hold.

$$C(T) \Leftrightarrow e \notin T \quad \forall e \in \mathcal{H}(\theta)$$

Why common screens fall short

Screen	Configuration	Blocker
SCR / gSCR	✓ local	✗
Modal	✓ local	✗
Additive	✓ approximate	✗
Pairwise	✓ approximate	incomplete
Full eigensolve $\mathcal{H}(\theta)$	✓ per case ✓ exact	enumerate all explicit set

9-candidate safe-set census: PENDING.

15/15
smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

$+0.127 \text{ s}^{-1}$
critical growth rate
canonical portfolio

0.6223 Hz
critical frequency
nominal IEEE-39 case

4 THE DEVICES REMAIN REGULAR; THE NETWORK CLOSES THE INSTABILITY

Exact factorization

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$
$$\det P_S(s) = h_S(s) \det T_0(s) \cdot \det(I + M_{SS}(s))$$
$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_{SS})$$
$$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$$

Devices to collective closure

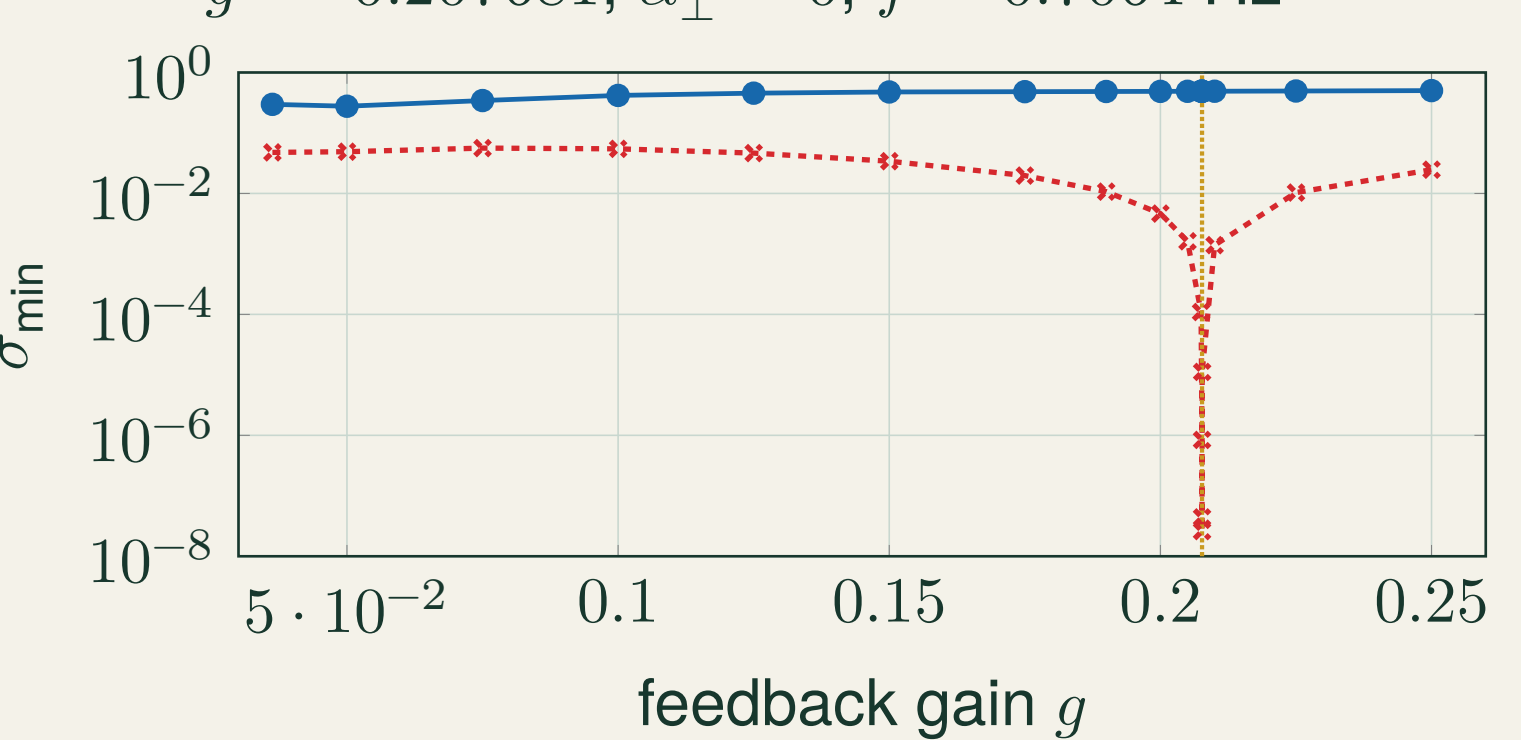
LOCAL DEVICE FACTORS
 $\prod_{i \in S} \det(I + M_{ii})$
each device remains regular

×

COLLECTIVE NETWORK CLOSURE
 $\det(I + Q_{SS})$
can close at zero

↓
network-mediated collective return
The local factors stay finite while the network return closes.

Boundary behavior



$g^* = 0.207681; \alpha_{\perp} = 0; f = 0.7064 \text{ Hz}$

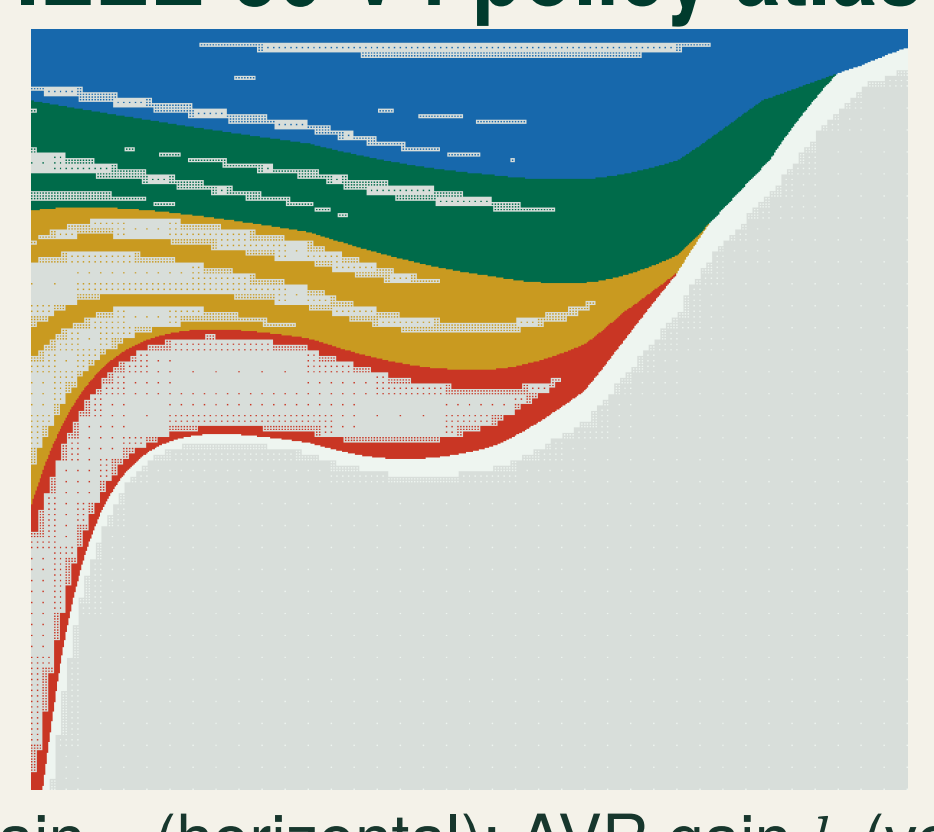
- local
- Q_H closure
- g^*

Every member is necessary in context
 H_4 is minimal; all proper subsets remain stable.

The devices remain locally regular.
Instability appears when the network-mediated collective return closes.
Strict M1 reduction: BLOCKED.
 $r = 2.34 \times 10^{-11}$; gate 1.00×10^{-12} .

5 THE SAME GRID CAN HAVE DIFFERENT MINIMAL BLOCKERS

IEEE-39 V4 policy atlas



Q/V gain g (horizontal); AVR gain k (vertical).
0 1 2 3 4 unsampled blocker order $|H_{\min}|$.

Distinct hypergraphs

30 36 16
F7A F7B F7C

Blocker-cardinality sequence

$$4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$$

One grid, different policy, different blocker structure.
Previously audited atlas; current policy campaign status: CONDITIONAL.

6 H_4 IS THE WITNESS — NOT THE UNIVERSAL BLOCKER

Frozen synthetic envelope (1000 scenario IDs)

967 valid (96.7%)

33 infeasible (3.3%) Scenario totals await review.

190/967
19.6%
 H_4 minimal

555/967
57.4%
any V_4 collective blocker

365/967
37.7%
alternative witnesses

PENDING USER REVIEW
Descriptive synthetic ensemble; IDs are not a probability sample. H_4 is minimal in the audited event.

7 A FIX THAT REPAIRS THE CANONICAL CASE IS NOT A UNIVERSAL CURE

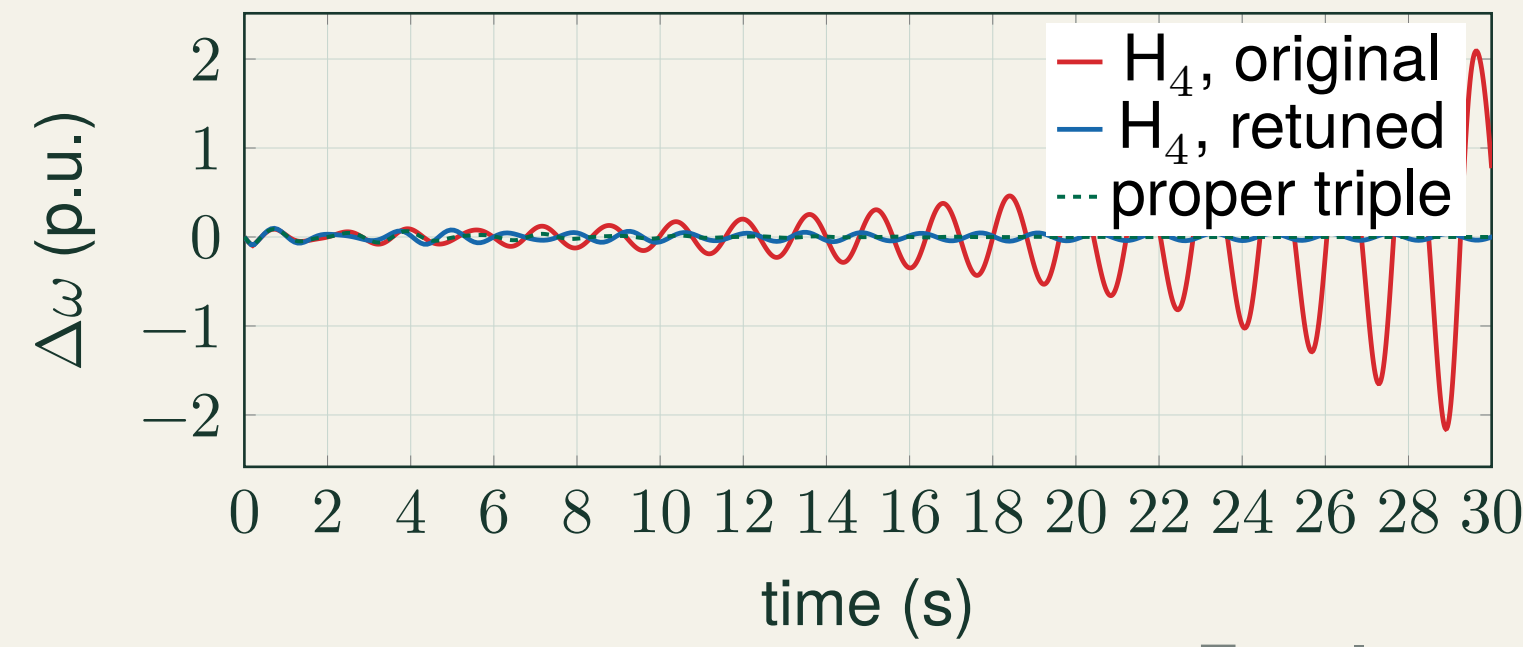
Control retuning: $g = 0.03625 \rightarrow 0.25$

594/967 61.4%
 $\alpha_{\perp}$ improves

186/967 19.2%
canonical H_4 rescued

373/967 38.6%
stability deteriorates

Canonical phasor-DAE response (0–30 s)



— H_4 , original
— H_4 , retuned
... proper triple

time (s)

Envelope: PENDING

8 AVOIDING EVERY HYPEREDGE PROTECTS THE WHOLE IMPLEMENTATION PATH

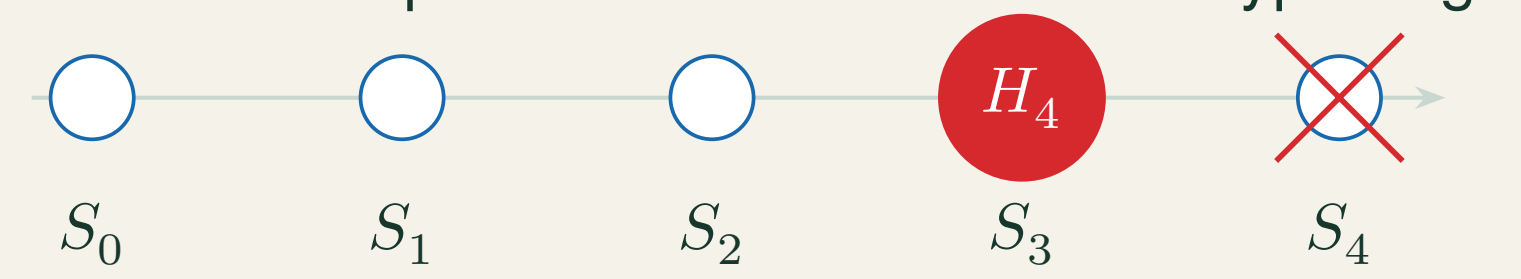
Path-constrained planning formulation

$$\max_S \sum_{i \in S} P_i \quad \text{s.t. } e \notin S, \quad \forall e \in \mathcal{H}(\theta)$$

Apply the same constraint to every intermediate prefix of the implementation path.

Unsafe implementation path

Schematic: the path reaches the canonical hyperedge.

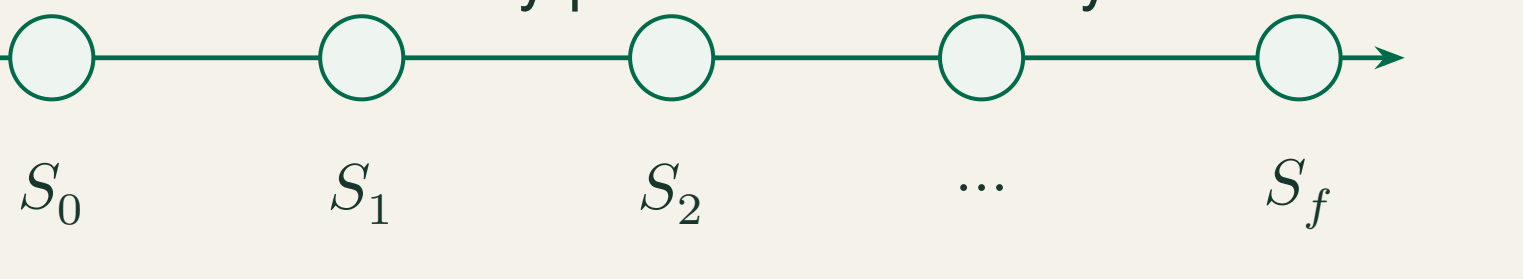


Reject the route when a prefix contains H_4 .

100/100 cross-code

Safe implementation path

Schematic: every prefix avoids every blocker.



Accept only after checking all path prefixes.

56/60 finite TDS agreement

P_{∞} PENDING
safe-set census pending

9 EVIDENCE IS MEASURED, SCOPED, AND READY TO AUDIT

100/100
same-model cross-code

56/60
finite TDS sign agreement

15/15
proper subsets stable

30/36/16
distinct policy blockers

594/967
improved; review pending

190/967
 H_4 minimal; review pending

Scope: phasor-domain DAE; stationary, re-equilibrated portfolios; no physical switching transient, EMT, current-limit, or DC-link claim; blocker identity depends on model and control policy.